MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

V SEMESTER EXAMINATION

(2024 ADMISION ONWARDS)

MG5DSCECT301 - Artificial Intelligence and Machine Learning

Duration: 2 hours Maximum Marks: 35

PRACTICAL EXAMINATION

Instructions:

• Demonstrate the given task. (Any one of the tasks listed below)
• Viva session (Minimum 5 questions from the Practical Module)
• Lab Report

Evaluation Criteria:

• Viva: 10 marks
• Lab Report: 5 marks
• Demonstration: 20 marks

Tasks: (Choose any one Program)

A. Understanding AI & Rational Agents
1. Simple Temperature Control Agent
2. Simulate a vacuum cleaner agent that cleans a dirty room.
3. create an agent that decides actions based on traffic signal colors.

B. Problem Formulation & Goal-Based Agent
4. Goal-Based Agent to Find an Item in a List
5. Goal-Based Agent to Reach a Destination
6. Goal-Based Number Guessing Agent

C. Greedy Best-First Search
7. Write a Python program to implement Greedy Best-First Search on a graph.
8. Write a Python program to implement A* search on a graph.

D. Hill Climbing Algorithm
9. Finding the Maximum in a List

10. Finding the largest element in an array by moving to neighbouring elements with higher value

E. Mini-Max Algorithm (Game AI)

11. Write a Python program to simulate a Stone Removal Game

F. Bayes Theorem Implementation

12. Calculate the probability of a student passing an examination given that the student has attended a particular number of classes.

G. Linear Regression (Basic ML Model)

13. Predict a student's marks based on the number of hours studied. Use a small dataset of study hours and marks.

H. K-Nearest Neighbours (KNN) Classification

14. Classify students into Pass or Fail based on their study hours and attendance percentage. Use K = 3.